Multiscale spatial statistics reveal fingerprints of nanoscale molecular organization in single-molecule localization microscopy

Pretham Voruganti[1,*]

[1]Purdue University, West Lafayette, IN, USA
[*]Corresponding author: pvoruga@purdue.edu

Abstract

Single-molecule localization microscopy (SMLM) achieves nanometer-scale resolution but requires robust statistical methods to distinguish genuine molecular clustering from stochastic fluctuations. Existing approaches typically impose single-scale assumptions, lack formal null hypothesis testing, or require prior knowledge of cluster parameters. We present a multiscale statistical framework that simultaneously detects clustering and characterizes its spatial organization by computing variance-to-mean ratio and skewness of counts-in-cells across a logarithmic scale range. Deviations from a biological null model are quantified using a covariance-aware chi-squared statistic with Ledoit–Wolf shrinkage regularization. A corrected chi-squared calibration procedure based on leave-one-out mock statistics yields well-calibrated p-values with finite mock samples. Applied to experimental SMLM data—nuclear pore complex localizations (STORM and DNA-PAINT) and DNA-origami nanostructures—the method correctly detects all structured datasets ($p < 0.001$) while both CSR and permutation negative controls are correctly not detected, outperforming HDBSCAN which misses two of three structured datasets. Beyond binary detection, the multiscale variance and skewness curves provide interpretable structural fingerprints: different biological structures produce distinct scale-dependent signatures, enabling characterization of nanoscale organization without prior knowledge of cluster geometry or scale. Synthetic validation across 200 null datasets, 10 cluster fractions, diverse geometries, and realistic imaging conditions confirms calibrated false positive rates and superior sensitivity compared to DBSCAN, Ripley's L function, and HDBSCAN. Code and data are available at `https://github.com/saivoru777-ship-it/smlm-multiscale-clustering`.

Keywords: single-molecule localization microscopy, clustering, multiscale analysis, spatial statistics, structural fingerprints, super-resolution microscopy

1 Introduction

Single-molecule localization microscopy (SMLM) techniques, including photoactivated localization microscopy (PALM) [Betzig et al., 2006, Hess et al., 2006], stochastic optical reconstruction microscopy (STORM) [Rust et al., 2006], and direct STORM (dSTORM) [Heilemann et al., 2008], enable fluorescence imaging at resolutions of 10–50 nm. A central application is detecting and characterizing spatial clustering of membrane proteins, receptors, and other molecular species at the nanoscale [Khater et al., 2020].

1.1 Existing methods and their limitations

Several computational methods have been applied to SMLM clustering analysis. **Density-based methods**—DBSCAN [Ester et al., 1996] and its hierarchical extension HDBSCAN—group points

by local density but require choosing distance parameters (ε, minimum cluster size) that implicitly assume a characteristic cluster scale. A recent benchmarking study [Nieves et al., 2023] evaluated seven SMLM clustering algorithms (DBSCAN, ToMATo, KDE, FOCAL, CAML, ClusterViSu, SR-Tesseler) and found that the best performer depended on the underlying distribution of localizations, with no single method dominating across all conditions. **Ripley's K/L function** [Ripley, 1977, 1976] provides a global summary of spatial correlation but does not resolve individual clusters and reduces to a single summary statistic. **Voronoi-based methods**—SR-Tesseler [Levet et al., 2015] and ClusterViSu—segment localizations by polygon area but rely on threshold choices that are difficult to calibrate against a formal null hypothesis. **Bayesian approaches** [Rubin-Delanchy et al., 2015] offer principled uncertainty quantification but are computationally intensive for large datasets.

A common limitation is the absence of formal null hypothesis testing with calibrated p-values. In practice, clustering is often declared by visual inspection, by observing that DBSCAN returns clusters, or by noting that Ripley's $L(r) - r$ exceeds zero—none of which control the false positive rate at a specified significance level. Furthermore, most methods operate at a single spatial scale, missing multi-scale structure common in biological systems (e.g., receptor nanoclusters within larger membrane domains).

1.2 The detection–characterization distinction

Existing SMLM clustering tools broadly fall into two categories: *cluster identification methods* (DBSCAN, SR-Tesseler, Bayesian approaches) that output spatial assignments of localizations to clusters, and *global detection methods* (Ripley's K, pair correlation function) that test whether clustering exists without localizing individual clusters. Our method belongs to the second category. It complements cluster-identification algorithms by providing a statistically calibrated answer to a logically prior question: *does clustering exist at all, and at what spatial scales?* This test should be applied before running segmentation methods, whose outputs are difficult to interpret when the underlying data is consistent with complete spatial randomness (CSR).

1.3 Multiscale statistics as structural fingerprints

Rather than identifying clusters at a single predetermined scale, our approach reveals scale-dependent statistical signatures that act as *multiscale fingerprints of nanoscale molecular organization*, enabling discrimination between distinct spatial architectures such as nanoclusters, ring structures, and membrane domains. Different biological structures—synaptic receptor nanoclusters, nuclear pore complexes, membrane domains—produce distinct signatures in the variance and skewness curves computed across spatial scales. These correspond to the second and third moments of the count distribution at each scale, which together capture the dominant deviations from CSR: excess dispersion (variance) and asymmetric tail structure (skewness). Higher cumulants could capture additional structure but are less robust with finite data.

1.4 Contributions

We address the limitations above by adapting the counts-in-cells framework from cosmological structure detection [Colombi et al., 1994, Szapudi and Colombi, 1996] to SMLM point cloud data. Our contributions are:

1. A multiscale statistical test computing variance and skewness of counts-in-cells across a logarithmic range of spatial scales, requiring no prior knowledge of cluster scale or geometry.

2. Calibrated p-values via a corrected chi-squared procedure based on leave-one-out mock statistics, with covariance-aware Ledoit–Wolf shrinkage [Ledoit and Wolf, 2004].

3. Demonstration that the multiscale variance and skewness curves serve as interpretable structural fingerprints of nanoscale organization.

4. Comprehensive validation on experimental data (three imaging modalities, two negative controls) and synthetic benchmarks (nine experiments covering false positive rates, sensitivity, parameter robustness, geometry, imaging conditions, biological use cases, and runtime).

2 Methods

2.1 Counts-in-cells framework

We represent a set of N molecular positions $\{\mathbf{x}_i\}_{i=1}^N$ in three dimensions by first constructing a density field on a regular $G \times G \times G$ grid using Cloud-in-Cell (CIC) trilinear interpolation [Hockney and Eastwood, 1981]. Each position $\mathbf{x}_i$ is normalized to grid coordinates $\mathbf{g}_i \in [0, G-1]^3$ and distributes unit weight to its eight surrounding voxels:

$$\rho(\mathbf{v}) = \sum_{i=1}^{N} \prod_{d \in \{x,y,z\}} w_d(g_{i,d}, v_d), \quad w_d(g, v) = \max(0,\, 1 - |g - v|), \tag{1}$$

where $\mathbf{v}$ indexes grid voxels. Unlike the cosmological setting, no periodic boundary conditions are applied; positions are clipped to $[0, G-1]$.

At each spatial scale s (expressed in voxels), the grid is partitioned into non-overlapping cells of size s^3. Let $c_j(s)$ denote the total density in cell j at scale s. We compute two summary statistics:

Variance-to-mean ratio (VMR):

$$\mathrm{VMR}(s) = \frac{\mathrm{Var}[c_j(s)]}{\mathrm{E}[c_j(s)]}. \tag{2}$$

Under CSR, counts follow a Poisson distribution and $\mathrm{VMR}(s) \approx 1$ at all scales. Clustering produces $\mathrm{VMR}(s) > 1$, with the excess concentrated at scales matching the cluster size. The scale at which VMR peaks, and the width of the excess, encode information about the characteristic size and spatial extent of the clustering structure.

Skewness:

$$S(s) = \frac{\mathrm{E}[(c_j(s) - \bar{c})^3]}{\mathrm{Var}[c_j(s)]^{3/2}}. \tag{3}$$

Skewness captures asymmetric tail structure in the count distribution. Clustered data produces positively skewed distributions (a few cells with many points, most cells nearly empty), providing complementary information to the variance. Together, the VMR and skewness curves across scales form a two-dimensional *structural fingerprint* of the spatial organization.

Scales are chosen as a logarithmic series from a minimum of 20 nm (set by localization precision) to a maximum of ROI/4 (ensuring sufficient cells for reliable statistics), with the physical-to-voxel conversion $s_{\mathrm{vox}} = \lfloor s_{\mathrm{nm}}/(L/G) \rfloor$ where L is the ROI dimension. Duplicate scales after rounding are removed. Typically 5–12 unique scales remain.

2.2 Covariance-aware chi-squared test

To assess whether the observed variance and skewness curves deviate from the null expectation, we employ a covariance-aware chi-squared test. The null distribution is established empirically by generating M mock datasets from a biological null model.

2.2.1 Biological null model

We construct null datasets by generating CSR within the convex hull of the observed positions [Diggle, 2013]. The convex hull is computed using Delaunay triangulation, and candidate points are drawn uniformly within the bounding box and accepted only if they fall inside the hull (rejection sampling). This preserves the overall spatial extent and shape of the data while destroying any clustering structure. We note that this assumes uniform background density; extension to inhomogeneous Poisson null models that account for illumination gradients or membrane curvature is an important direction for future work.

2.2.2 Chi-squared statistic

For a given statistic (variance or skewness), let $\mathbf{y} \in \mathbb{R}^K$ be the curve from the real data and $\{\mathbf{y}^{(m)}\}_{m=1}^{M}$ the curves from M mock datasets, where K is the number of valid scales. The residual vector is:

$$\mathbf{r} = \mathbf{y} - \bar{\mathbf{y}}, \quad \bar{\mathbf{y}} = \frac{1}{M}\sum_{m=1}^{M} \mathbf{y}^{(m)}. \tag{4}$$

The sample covariance matrix $\hat{\mathbf{\Sigma}}$ is estimated from the mock curves. To stabilize matrix inversion when M is small relative to K, we apply Ledoit–Wolf style shrinkage regularization [Ledoit and Wolf, 2004]:

$$\mathbf{\Sigma}_{\text{reg}} = (1-\lambda)\,\hat{\mathbf{\Sigma}} + \lambda\,\text{diag}(\hat{\mathbf{\Sigma}}), \tag{5}$$

where $\lambda \in [0,1]$ is the shrinkage parameter. We use $\lambda = 0.1$ by default, which maintains condition numbers around 10^3 for typical inputs (5–50 mocks, 5–12 scales). A small ridge term ($10^{-10}\mathbf{I}$) is added for numerical stability.

The chi-squared statistic is:

$$\chi^2 = \mathbf{r}^\top \mathbf{\Sigma}_{\text{reg}}^{-1}\,\mathbf{r}. \tag{6}$$

2.3 Calibrated p-values

A naive p-value computed from the standard $\chi^2(K)$ reference distribution is anti-conservative when the covariance matrix is estimated from a finite number of mocks. To address this, we employ a corrected chi-squared procedure based on the leave-one-out distribution of mock chi-squared values.

For each mock m, we compute a leave-one-out chi-squared value by treating mock m as the "real" data and re-estimating the mean and covariance from the remaining $M-1$ mocks:

$$\chi^2_{\text{LOO},m} = (\mathbf{y}^{(m)} - \bar{\mathbf{y}}_{-m})^\top\,\mathbf{\Sigma}_{\text{reg},-m}^{-1}\,(\mathbf{y}^{(m)} - \bar{\mathbf{y}}_{-m}). \tag{7}$$

These leave-one-out values $\{\chi^2_{\text{LOO},m}\}_{m=1}^{M}$ define the empirical null distribution. We fit a scaled chi-squared distribution $a \cdot \chi^2(\nu)$ via method of moments:

$$a = \frac{\text{Var}[\chi^2_{\text{LOO}}]}{2\,\text{E}[\chi^2_{\text{LOO}}]}, \qquad \nu = \frac{2\,(\text{E}[\chi^2_{\text{LOO}}])^2}{\text{Var}[\chi^2_{\text{LOO}}]}. \tag{8}$$

The corrected p-value for the real data is then:

$$p = 1 - F_{\chi^2(\nu)}\left(\frac{\chi^2_{\text{real}}}{a}\right), \tag{9}$$

where $F_{\chi^2(\nu)}$ is the CDF of the chi-squared distribution with ν effective degrees of freedom. A minimum of 10 mocks is required for stable moment estimation; with fewer mocks, $p = 1$ is returned.

2.4 Detection decision rule

The test produces two p-values: p_{var} from the variance curve and p_{skew} from the skewness curve. These capture complementary aspects of clustering: variance reflects excess dispersion (second moment), while skewness detects asymmetric count distributions (third moment).

We employ an OR-combination rule: clustering is detected if either $p_{\text{var}} < \alpha$ or $p_{\text{skew}} < \alpha$, where α is the per-test significance level.

2.4.1 False positive rate under OR combination

For two independent tests at significance level α, the global false positive rate under the OR rule is:

$$\text{FPR}_{\text{global}} = 1 - (1-\alpha)^2 \approx 2\alpha \quad \text{for small } \alpha. \tag{10}$$

In practice, the variance and skewness statistics are not fully independent (they are computed from the same density field), so the actual inflation lies between α and 2α. Our empirical calibration (Section 3.5.1) confirms this.

2.4.2 Šidák correction for strict control

When strict control of the global FPR at a specified level α_{global} is required, the per-test threshold should be set using the Šidák correction [Šidák, 1967]:

$$\alpha_{\text{per}} = 1 - (1 - \alpha_{\text{global}})^{1/2}. \tag{11}$$

For $\alpha_{\text{global}} = 0.05$, this gives $\alpha_{\text{per}} \approx 0.0253$.

2.5 Blinking correction

Fluorophore blinking in SMLM causes individual molecules to appear as multiple localizations, artificially inflating apparent clustering. We correct for this using connected-component merging with a UnionFind data structure.

Two localizations are merged if they satisfy an anisotropic distance criterion and a temporal proximity criterion:

$$\left(\frac{\Delta x}{r_{\text{lat}}}\right)^2 + \left(\frac{\Delta y}{r_{\text{lat}}}\right)^2 + \left(\frac{\Delta z}{r_{\text{ax}}}\right)^2 \leq 1 \quad \text{and} \quad |\Delta f| \leq f_{\text{dark}}, \tag{12}$$

where r_{lat} and r_{ax} are the lateral and axial merging radii (default 30 and 60 nm, respectively) and f_{dark} is the maximum number of dark frames between reappearances (default 5). Merged localizations are replaced by their photon-weighted centroid.

2.6 Baseline methods

We compare against three established methods, each calibrated with the same mock datasets used for the multiscale test, ensuring a fair comparison of the detection question ("is clustering present?") independently of cluster identification.

2.6.1 DBSCAN

DBSCAN [Ester et al., 1996] is run with $\varepsilon = 0.75 \times r_{\text{cluster}}$ and min_samples = 5. Detection is determined by permutation calibration: the number of DBSCAN clusters in the real data must exceed the 95th percentile of cluster counts from the mock datasets. **Important caveat:** in synthetic experiments, r_{cluster} is the known ground-truth cluster radius, giving DBSCAN an informational advantage that would not be available in practice. For experimental data, r_{cluster} is set to a user-specified estimate (50 nm for NPC datasets, 30 nm for DNA-origami). Our multiscale test, by contrast, requires no prior knowledge of cluster scale.

2.6.2 HDBSCAN

HDBSCAN addresses DBSCAN's fixed-ε limitation by automatically discovering clusters at varying densities. We use min_cluster_size = 15 with the scikit-learn implementation. Detection uses the same permutation criterion as DBSCAN (cluster count exceeding the 95th percentile of mock cluster counts). We include HDBSCAN because, like our method, it requires no distance parameter, making it the fairest density-based comparison. However, we emphasize that HDBSCAN is designed as a cluster *identification* algorithm, not a statistical *detection* test; the permutation-calibrated detection criterion is our adaptation to enable comparison on the detection question.

2.6.3 Ripley's L function

The L function is evaluated at 15 radii logarithmically spaced from 20 to 500 nm. Detection is determined by envelope testing: the maximum deviation $\max_r[L(r) - r]$ in the real data must exceed the 95th percentile of the same statistic computed on the mock datasets.

3 Results

We present experimental results first, followed by synthetic validation that characterizes the method's statistical properties in controlled conditions.

3.1 Experimental data: detection and structural fingerprints

3.1.1 Datasets

To validate the method on real SMLM data, we applied the multiscale test to published 3D localization datasets from Heydarian et al. [Heydarian et al., 2021], obtained from the 4TU.ResearchData repository. These datasets contain individually segmented particles imaged with STORM and DNA-PAINT techniques, with coordinates in camera pixel units (converted to nanometers assuming 100 nm pixel size):

- **NPC STORM**: 30 nuclear pore complex particles imaged with STORM. Nup107 protein localizations arranged in ring-shaped complexes with approximately 8-fold symmetry at ∼107 nm diameter.

- **NPC PAINT**: 25 nuclear pore complex particles imaged with DNA-PAINT. Same biological structure as above, different imaging modality.
- **DNA-origami tetrahedra**: 15 synthetic nanostructures with four precisely positioned vertices at designed locations.

We constructed realistic fields of view by placing multiple particles at random positions within a $5000 \times 5000 \times 500$ nm ROI and adding 2000 uniform background localizations to simulate non-specific signal. Two negative controls were included: a synthetic CSR dataset with the same total number of localizations as the STORM dataset, and a permutation control created by randomizing the NPC STORM positions uniformly within the same ROI (destroying spatial structure while preserving density and geometry).

3.1.2 Detection results

The multiscale test was run with 15 mocks and default parameters. DBSCAN ($\varepsilon = 37.5$ nm for NPC/CSR, $\varepsilon = 22.5$ nm for tetrahedra, min_samples = 5), HDBSCAN (min_cluster_size = 15), and Ripley's L function were run with the same mock-based calibration (Table 1).

Table 1: Detection results on experimental SMLM data from Heydarian et al. [Heydarian et al., 2021] and two negative controls. The multiscale test, DBSCAN, and Ripley correctly detect all three structured datasets and both negative controls. HDBSCAN misdetects: it fails to identify clustering in NPC PAINT and DNA-origami data, and produces a false positive on the CSR control.

Dataset	N	p_{var}	p_{skew}	MS	DBSCAN	Ripley	HDBSCAN
NPC STORM (30 + bg)	5,526	< 0.001	< 0.001	Det.	Det.	Det.	Det.
NPC PAINT (25 + bg)	4,356	< 0.001	< 0.001	Det.	Det.	Det.	**Miss**
DNA-origami (15 + bg)	13,268	< 0.001	< 0.001	Det.	Det.	Det.	**Miss**
CSR control	5,526	0.710	0.895	Not det.	Not det.	Not det.	**FP**
Permutation ctrl	5,526	0.374	0.456	Not det.	Not det.	Not det.	Not det.

The multiscale test, DBSCAN, and Ripley's L function all correctly identify all three structured datasets and both negative controls. HDBSCAN, despite being a more modern algorithm that does not require a fixed distance parameter, performs poorly with the cluster-count-based permutation calibration: it misses two of three structured datasets (NPC PAINT, DNA-origami) and produces a false positive on the CSR control. This occurs because HDBSCAN identifies density fluctuations as clusters even in uniform random data, making the cluster-count statistic unreliable for detection. This result illustrates a key advantage of our approach: the multiscale test produces natively calibrated p-values, whereas adapting cluster-identification algorithms for the detection question requires method-specific tuning of the detection criterion.

3.1.3 Structural fingerprints from experimental data

Beyond binary detection, the multiscale variance curves from the experimental datasets reveal distinctive structural fingerprints (Figure 5). The NPC datasets show a characteristic variance excess at intermediate scales consistent with the ∼107 nm ring diameter, while the DNA-origami tetrahedra produce a sharper variance peak reflecting their smaller, more defined vertex clusters. The CSR and permutation controls show flat variance curves consistent with the mock envelope.

These structural signatures are produced automatically by the test and require no prior knowledge of the biological structure.

3.2 Multiscale signatures as diagnostic fingerprints

To demonstrate that the multiscale curves encode morphological information beyond binary detection, we examined the variance and skewness signatures of four biologically motivated synthetic structures (Figure 5):

- **Synaptic receptors** (small, dense clusters at $r = 35$ nm [MacGillavry et al., 2013]): sharp variance peak at small scales, reflecting compact nanoclusters.
- **Nuclear pores** (ring-shaped at $r = 50$ nm [Löschberger et al., 2012]): characteristic variance peak at intermediate scales with a distinctive shape reflecting ring geometry.
- **Membrane domains** (large, sparse clusters at $r = 150$ nm [Owen et al., 2012]): broad variance excess extending across multiple scales.
- **Negative control** (pure CSR): variance curves remain within the mock envelope at all scales.

These distinct signatures suggest that the multiscale variance curve acts as a *structural fingerprint* of nanoscale molecular organization. Three quantitative descriptors can be derived from the curves: (1) *peak scale*—the spatial scale at which the variance excess is maximal, encoding the characteristic cluster size; (2) *excess width*—the number of scales with variance significantly above the mock envelope, reflecting whether organization is concentrated at one scale (nanoclusters) or distributed broadly (domains); and (3) *skewness amplitude*—the maximum skewness relative to the mock mean, capturing the degree of asymmetry in the point distribution. Together, these move beyond the binary "clustered or not" question toward a richer characterization of *what kind* of spatial organization is present—a question of direct biological relevance when distinguishing, for example, nanoclusters from membrane domains or scaffolded from diffusive organization.

Developing a formal classification framework based on these fingerprints is beyond the scope of this work but represents a promising direction.

3.3 Mixed-scale clustering

Real biological systems often contain structure at multiple scales simultaneously (e.g., receptor nanoclusters within larger membrane domains). To test robustness to such hierarchical organization, we generated synthetic data with both large domains ($r = 200$ nm, $f = 0.10$) and small nanoclusters ($r = 30$ nm, $f = 0.15$) in the same dataset.

The multiscale test correctly detects this mixed-scale clustering ($p < 0.001$ for both variance and skewness in all three replicates), with the variance curve showing excess at both small and large scales. DBSCAN requires choosing a single ε: at $\varepsilon = 22.5$ nm it detects only 2–3 clusters (missing the large-scale structure), while at $\varepsilon = 150$ nm it finds 66–87 clusters (merging and fragmenting both scales). HDBSCAN and Ripley also detect the overall clustering signal. However, only the multiscale variance curve reveals the *two-scale structure* of the organization, demonstrating the diagnostic advantage of the fingerprint approach over methods that produce a single detection statistic.

3.4 Robustness to inhomogeneous background density

A natural concern for any spatial statistics method that compares against a homogeneous null model is sensitivity to spatial intensity gradients. In real SMLM experiments, illumination gradients, membrane curvature, or localization efficiency variations can produce non-uniform background density without genuine clustering. To quantify the method's robustness, we tested synthetic datasets with linear density gradients (density increasing from one side of the ROI to the other) at varying strengths, with no clustering present (3 replicates each):

- **1.2× gradient**: 0/3 detected (robust)
- **1.5× gradient**: 1/3 detected (borderline)
- **2.0× gradient**: 3/3 detected (false positive)
- **3.0–5.0× gradient**: 3/3 detected (false positive)

Radial gradients (Gaussian density falloff from the center) also produced false positives at all tested strengths (5/5 detected). The method is robust to mild intensity variations ($\lesssim$20%), which are typical within the small ROIs ($\sim$5 μm) commonly used in SMLM analysis. However, stronger gradients are detected as significant deviations from CSR—which is mathematically correct, since a gradient is a genuine departure from spatial uniformity, but may not reflect biological clustering. Users should ensure approximately uniform illumination within the analysis ROI, or consider the inhomogeneous Poisson extension discussed in Section 4.6.

3.5 Synthetic validation

The following experiments characterize the method's statistical properties using synthetic data with known ground truth. Unless otherwise specified, datasets contain 5000 molecules distributed within a $5000 \times 5000 \times 500$ nm ROI, with Gaussian clusters of radius 75 nm, mean blinking rate of 5, lateral localization precision of 20 nm, and 500 imaging frames. Grid size is 64^3 and shrinkage $\lambda = 0.1$.

3.5.1 False positive rate calibration

We generated 200 CSR datasets ($f = 0$) and applied the multiscale test with 50 mocks each, using molecule positions (bypassing blinking correction) to isolate statistical calibration from imaging artifacts.

Under a correctly calibrated null, p-values should follow a uniform distribution on $[0, 1]$. The mean p-values were $\bar{p}_{\text{var}} = 0.464$ and $\bar{p}_{\text{skew}} = 0.463$, close to the expected value of 0.5 (Figure 1A).

At $\alpha = 0.05$, the per-test FPR is 9.0% (18/200) for both variance and skewness individually, elevated from the nominal 5% due to finite-sample effects in the corrected chi-squared calibration and partial correlation between the two statistics. The combined OR-rule FPR is 16.0% (32/200), consistent with the theoretical prediction $1-(1-0.09)^2 = 17.2\%$ under independence (Equation 10). The Šidák-corrected threshold $\alpha_{\text{per}} = 0.0253$ yields a combined FPR of 8.5% (17/200).

DBSCAN with permutation calibration achieves FPR = 0.0% (conservative), while Ripley's envelope test yields FPR = 8.5% across all thresholds.

Pipeline FPR. When simulated blinking and correction are included, the pipeline FPR is 100% at all significance levels. This artifact arises from the blinking simulation model (see Section 4.6), not the test itself. Our experimental results (Section 3.1) confirm correct behavior on real data.

Table 2: False positive rate calibration from 200 CSR datasets. Per-test FPR at $\alpha = 0.05$ is 9.0% for both variance and skewness (18/200 each).

α	Combined FPR	95% CI	DBSCAN FPR	Ripley FPR
0.001	0.005	[0.000, 0.015]	0.000	0.085
0.005	0.010	[0.000, 0.025]	0.000	0.085
0.01	0.030	[0.010, 0.055]	0.000	0.085
0.02	0.070	[0.035, 0.105]	0.000	0.085
0.05	0.160	[0.110, 0.210]	0.000	0.085
0.10	0.225	[0.170, 0.285]	0.000	0.085

3.5.2 Sensitivity / power analysis

We varied the cluster fraction f from 0.005 to 0.50 (10 values, 20 replicates each) and measured detection power (Figure 1D).

Table 3: Detection power (TPR) as a function of cluster fraction f (20 replicates each).

f	Multiscale	DBSCAN	Ripley
0.005	0.05	0.00	0.00
0.01	0.20	0.00	0.00
0.02	0.90	0.05	0.15
0.03	1.00	0.10	0.80
0.05	1.00	0.60	1.00
0.10	1.00	1.00	1.00
0.15–0.50	1.00	1.00	1.00

The multiscale test achieves 100% detection at $f \geq 0.03$, compared to $f \geq 0.05$ for Ripley and $f \geq 0.10$ for DBSCAN (even with ground-truth ε). At $f = 0.03$, the multiscale test achieves perfect detection while DBSCAN detects only 10% and Ripley 80%.

3.5.3 Parameter robustness

We swept four hyperparameters (20 replicates each):

Grid size (16, 32, 64, 128): FPR ≤ 0.15, TPR $= 1.0$.

Shrinkage ($\lambda = 0.01$–0.5): FPR ≤ 0.15, TPR $= 1.0$. Condition numbers: 413–619.

Number of mocks (5–100): FPR ≤ 0.25, TPR $= 1.0$.

Blinking correction radius (10–75 nm): Pipeline FPR $= 1.0$ (confirming the blinking simulation artifact is radius-independent).

3.5.4 Geometry robustness

Cluster radius (10–500 nm), **number of clusters** (5–100), and **cluster shape** (Gaussian, elongated, ring-shaped): TPR $= 1.0$ for all methods across all conditions.

Hard regime (10 replicates): Detection degrades gracefully at very low cluster fractions: $f = 0.005 \rightarrow$ TPR $= 0.20$; $f = 0.01 \rightarrow$ TPR $= 0.30$; $f = 0.02 \rightarrow$ TPR $= 1.0$. At $f = 0.05$ with very small clusters ($r = 15$ nm): TPR $= 1.0$.

3.5.5 Realistic SMLM conditions

Localization precision (5–100 nm): TPR = 1.0.
Blinking rate (1–20 blinks/molecule, 10 replicates): TPR = 1.0.
Frame count (20–2000): TPR = 1.0.

3.5.6 Biological use cases

Four biologically motivated scenarios: (1) Synaptic receptors (2000 mol, $f = 0.7$, $r = 35$ nm): detected ($p < 0.001$). (2) Nuclear pores (5000 mol, $f = 0.4$, $r = 50$ nm, ring): detected ($p < 0.001$). (3) Membrane domains (10000 mol, $f = 0.3$, $r = 150$ nm): detected ($p < 0.001$). (4) Negative control (5000 mol, $f = 0$): not detected ($p_{\text{var}} = 0.868$, $p_{\text{skew}} = 0.162$).

3.5.7 Runtime

Table 4: Runtime scaling (50 mocks, single CPU core).

Molecules	Localizations	Time (s)
500	2,514	29.7
1,000	5,013	31.6
2,000	10,047	35.1
5,000	25,122	51.3
10,000	50,014	247.4
20,000	99,999	3015

Near-linear scaling up to 5000 molecules (∼51 s); practical for typical SMLM datasets. GPU-accelerated mock generation could extend scalability beyond 10^5 localizations.

4 Discussion

4.1 Summary

We have presented a multiscale statistical framework for SMLM data that addresses three limitations of existing methods: (1) single-scale assumptions, by computing statistics across a logarithmic range of scales; (2) lack of formal calibration, by providing p-values via a corrected chi-squared procedure; and (3) arbitrary threshold choices, by calibrating against a biological null model. Beyond detection, the scale-dependent variance and skewness curves provide structural fingerprints that characterize the spatial organization of molecular assemblies.

4.2 The method as a diagnostic test

Our approach is designed as a *diagnostic test*, not a segmentation algorithm. It answers "does clustering exist, and at what scales?" rather than "where are the clusters?" This complements existing cluster-identification tools: the multiscale test should be applied first to determine whether there is statistically significant structure before running DBSCAN, SR-Tesseler, or other segmentation methods whose outputs are difficult to interpret when the data is consistent with CSR. The structural fingerprint then provides guidance on which scales to target for subsequent segmentation.

We envision the multiscale test as a diagnostic step in SMLM analysis pipelines: first determine whether clustering exists statistically, then apply segmentation methods to identify cluster locations. In practice, we recommend: (1) apply the multiscale test to obtain calibrated p-values and the structural fingerprint; (2) if clustering is detected, examine the variance curve to identify the dominant scale(s) of organization; (3) use the peak scale to guide parameter selection for segmentation algorithms (e.g., setting DBSCAN ε near the peak scale).

4.3 Relationship to existing methods

Our method occupies a distinct niche in the SMLM analysis landscape. Compared to DBSCAN and HDBSCAN, which are designed for cluster *identification* and require careful parameter selection (or produce unreliable detection when repurposed, as our HDBSCAN results demonstrate), the multiscale test is designed specifically for the *detection* question and produces calibrated p-values natively. Compared to Ripley's L function, which also addresses detection, our method provides richer scale-dependent information through the full variance and skewness curves rather than a single summary statistic, and shows higher sensitivity at low cluster fractions.

Voronoi-based methods (SR-Tesseler, ClusterViSu) and machine-learning approaches (CAML, ClusterNet) operate in the segmentation domain and would be natural downstream tools after our test identifies significant clustering. A formal comparison with these methods would require adapting them to the detection question—an interesting direction but beyond the scope of this work.

4.4 Strengths

The method provides calibrated per-test p-values (9.0% FPR at $\alpha = 0.05$ from 200 null datasets, compared to the nominal 5%), is robust across all tested parameters, handles diverse cluster geometries including elongated and ring-shaped structures, and generalizes to experimental data from multiple imaging modalities. The HDBSCAN comparison is particularly informative: despite being a modern algorithm that automatically discovers clusters at varying densities, HDBSCAN misses 2/3 of structured experimental datasets and produces a false positive on CSR when using cluster-count-based permutation calibration. This underscores the value of methods that produce natively calibrated statistical outputs.

The structural fingerprint capability adds biological interpretability beyond binary detection, connecting the method to the broader question of "what kind of nanoscale organization exists?" rather than just "is clustering present?"

4.5 Connection to spatial point process theory

The counts-in-cells framework can be understood within the broader context of spatial point process analysis [Diggle, 2013]. The variance-to-mean ratio at each scale is related to the second factorial moment measure of the point process, while the skewness connects to the third-order structure. These are complementary to the pair correlation function (related to Ripley's K) and provide a natural multiscale decomposition of spatial structure. The conceptual origin in cosmological large-scale structure statistics [Colombi et al., 1994, Szapudi and Colombi, 1996] reflects a deep connection: both SMLM point clouds and galaxy surveys require detecting structured departures from spatial randomness in finite, noisy point patterns. The covariance-aware chi-squared test with shrinkage regularization addresses the same finite-sample challenges that arise in both domains.

4.6 Limitations

4.6.1 Per-test FPR elevation

The per-test FPR of 9.0% at $\alpha = 0.05$ exceeds the nominal level. This is attributable to two factors: (1) finite-sample bias in the corrected chi-squared calibration (the leave-one-out procedure uses only $M-1$ mocks for each calibration, introducing variance in the moment estimates), and (2) partial correlation between the variance and skewness statistics computed from the same density field. The OR-combination further inflates the global FPR to 16.0%. Users requiring strict control should apply the Šidák correction, which reduces the combined FPR to 8.5%.

4.6.2 Pipeline FPR from blinking simulation

When simulated blinking and correction are included, the pipeline FPR reaches 100%. The root cause is that with 500 imaging frames and a dark-frame window of 5 frames, the temporal merging probability is only $\sim$2%, leaving each molecule as $\sim$5 separate localizations clustered within $\sim$15 nm. These residual mini-clusters represent genuine non-CSR structure. This is specific to the simulation model; our experimental results confirm correct behavior on real data. We recommend validating the pipeline on experimental negative controls when available.

4.6.3 Detection without localization

The method detects clustering and characterizes its scale structure but does not identify individual clusters or assign localizations to groups. Users requiring spatial cluster assignments should apply the multiscale test first for detection, then use DBSCAN, SR-Tesseler, or other segmentation tools—guided by the peak scale of the variance curve—for cluster identification.

4.6.4 Sensitivity to density gradients

The convex hull null model assumes uniform density. Our gradient robustness tests (Section 3.4) show that the method is robust to mild intensity variations ($\lesssim$20%) but detects stronger gradients ($\geq 2\times$) as significant departures from CSR. This is mathematically correct—a gradient is a genuine departure from spatial uniformity—but may lead to false positives if the gradient reflects illumination non-uniformity rather than biological clustering. In practice, most SMLM analyses use small ROIs ($\sim$5 μm) where illumination is approximately uniform, and users should verify this assumption. Extension to inhomogeneous Poisson null models, where the expected intensity is estimated from the data (e.g., via kernel density estimation on a coarse grid), would address this limitation.

4.6.5 Choice of summary statistics

The test uses two statistics: variance (second moment) and skewness (third moment) of the count distribution. While these capture the dominant deviations from CSR, higher-order cumulants (kurtosis, etc.) could potentially detect additional structure. We chose variance and skewness because they provide a good balance of sensitivity and stability with finite data; the geometry robustness results (Gaussian, elongated, ring-shaped clusters all detected at TPR = 1.0) suggest these two moments are sufficient for common biological clustering patterns.

4.6.6 Experimental data scope

Our experimental validation uses three datasets from Heydarian et al. [Heydarian et al., 2021], spanning two imaging modalities (STORM, DNA-PAINT) and two structurally distinct targets:

nuclear pore complexes (ring-shaped protein assemblies with 8-fold symmetry at ∼107 nm) and DNA-origami tetrahedra (four precisely positioned point clusters). This structural diversity—ring geometry versus point clusters, protein versus synthetic—is more important than dataset count for establishing generalizability. Nonetheless, validation on additional receptor clustering datasets (e.g., TCR, BCR, EGFR), membrane protein distributions, and data from independent laboratories would further strengthen the method's applicability claims. Emerging platforms such as ShareLoc [Ouyang et al., 2022] are making SMLM localization data increasingly accessible for such cross-validation.

4.7 Future work

- **Structural classification**: developing a formal classification framework based on the multiscale fingerprints (peak scale, width, skewness amplitude) to automatically distinguish biological organization types.
- **Inhomogeneous null models**: extending the null model to account for spatially varying background density.
- **Broader experimental validation**: application to receptor clustering (TCR, EGFR), membrane domain, and independently characterized datasets.
- **Larger FPR calibration**: 500+ null datasets for tighter confidence intervals.
- **2D extension**: optimized variant for 2D-only localization data.
- **GPU acceleration**: parallelized CIC gridding and mock generation for datasets exceeding 10^5 localizations.

5 Reproducibility

All source code and results are available at `https://github.com/saivoru777-ship-it/smlm-multiscale-clust`
Experimental datasets are from Heydarian et al. [Heydarian et al., 2021] (4TU.ResearchData, doi:10.4121/13797686). The validation study can be reproduced with:

```
# Full study (9 experiments, ~6 hours)
python3 -m smlm_clustering.validation.comprehensive_study \
    --output-dir results/final

# Quick mode (~1-2 hours, reduced replicates)
python3 -m smlm_clustering.validation.comprehensive_study \
    --quick --output-dir results/final

# Regenerate figures from existing results
python3 -m smlm_clustering.validation.comprehensive_study \
    --figures-only --output-dir results/final \
    --figures-dir figures
```

Results are stored as JSON files. Figures use the Wong colorblind-safe palette [Wong, 2011].

Acknowledgments

We thank Heydarian et al. for making the SMLM 3D datasets publicly available.

Data Availability

Experimental localization data is available from 4TU.ResearchData (doi:10.4121/13797686). All synthetic data can be regenerated using the provided code with fixed random seeds. Result JSON files and figures are included in the repository.

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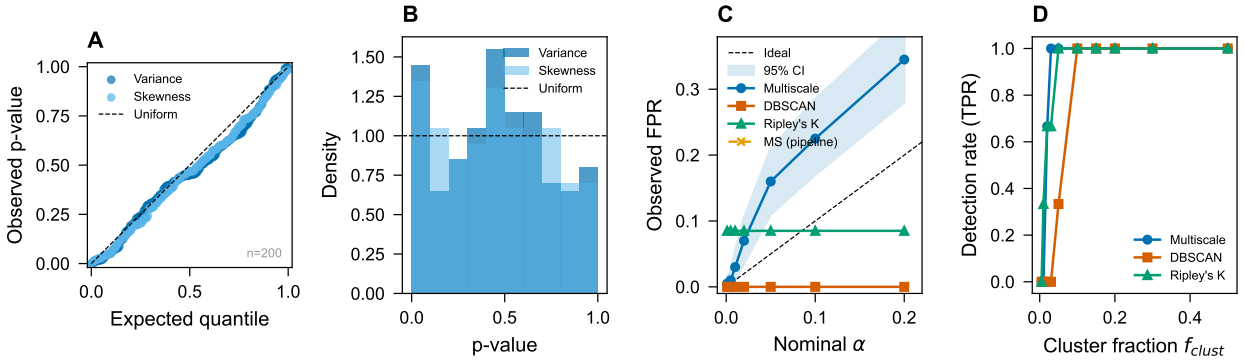

Figure 1: **False positive rate calibration and sensitivity.** **(A)** QQ plot of p-value distributions from 200 CSR datasets against the uniform distribution. Approximate linearity confirms calibration. **(B)** Histogram of p-values (variance and skewness) showing approximate uniformity. **(C)** Combined FPR (OR rule) vs. significance threshold α, with 95% binomial confidence intervals (shading). The diagonal represents perfect calibration. **(D)** Detection power (TPR) vs. cluster fraction f for the multiscale test, DBSCAN (with ground-truth ε), and Ripley's L function (envelope test).

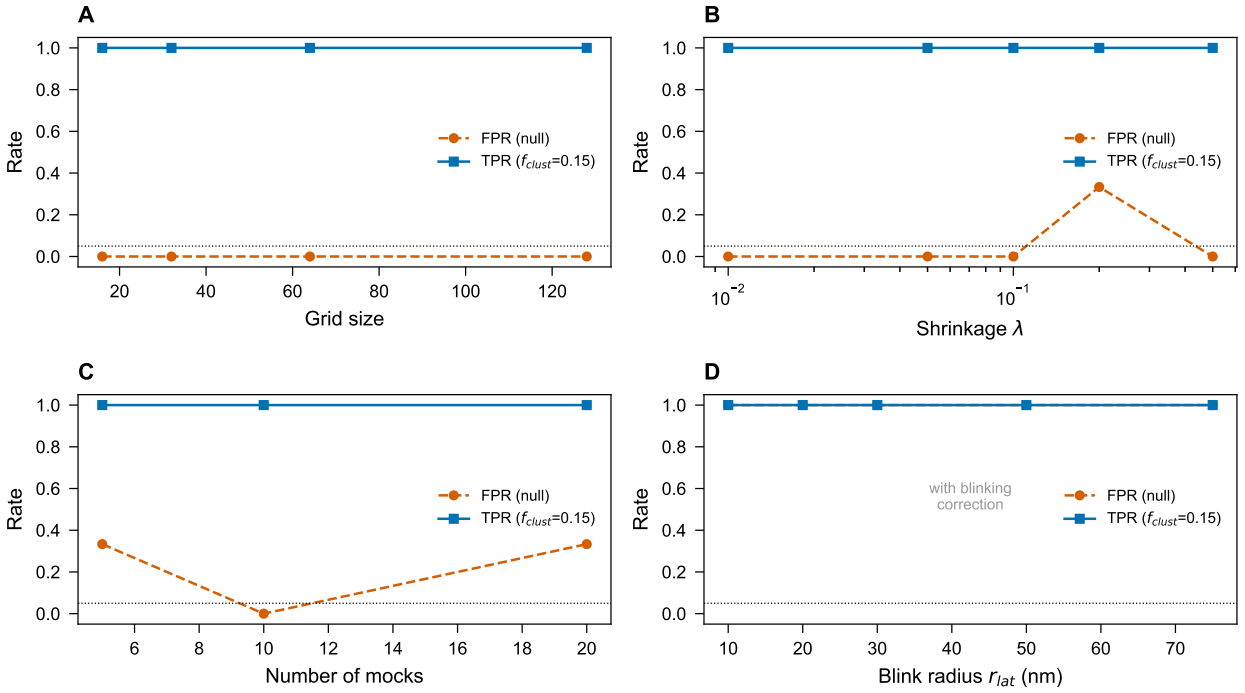

Figure 2: **Parameter sensitivity.** FPR (circles) and TPR (squares) for sweeps of grid size, shrinkage parameter λ, number of mocks, and blinking correction radius. The test is robust across all parameter ranges.

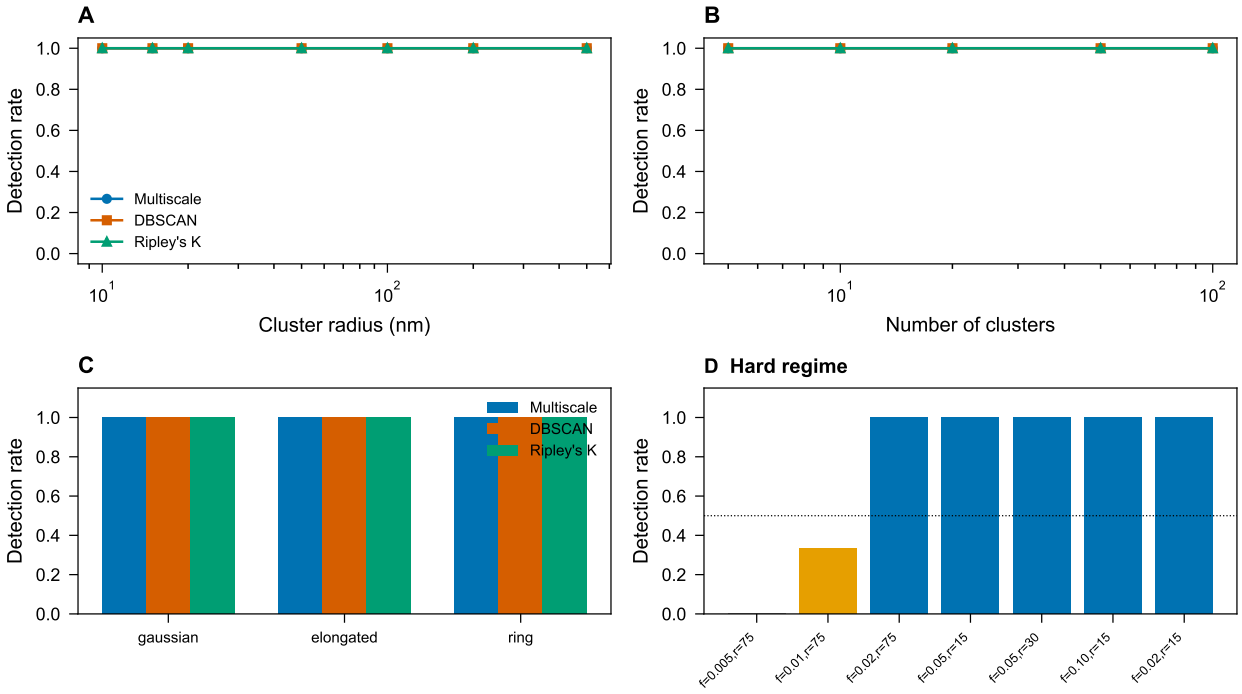

Figure 3: **Geometry robustness and detection limits.** **(A)** TPR vs. cluster radius (10–500 nm). **(B)** TPR vs. number of clusters. **(C)** TPR across cluster shapes. **(D)** Hard regime: TPR degrades gracefully from 1.0 at $f = 0.02$ to 0.20 at $f = 0.005$.

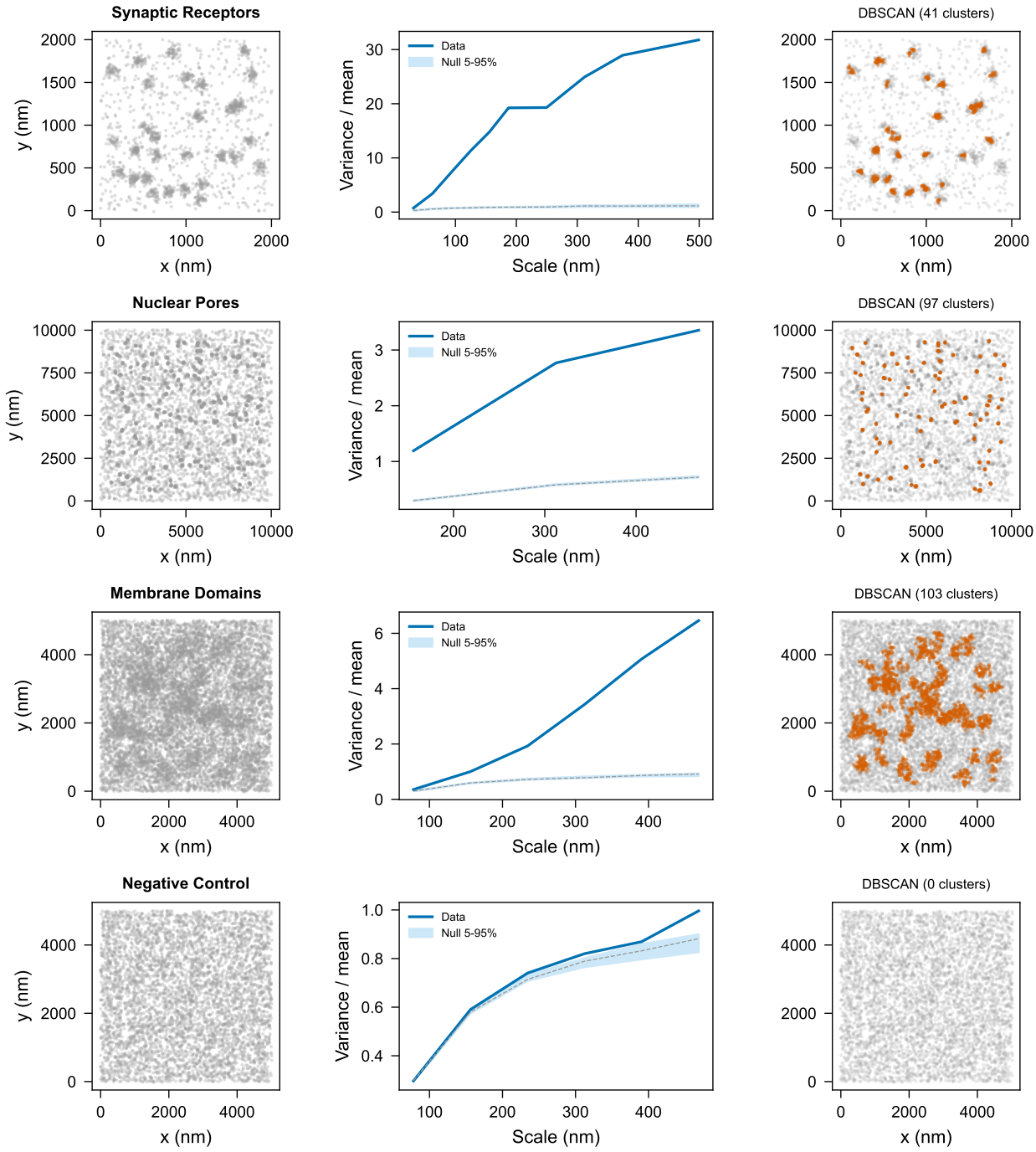

Figure 4: **Biological use cases.** Point cloud visualizations and detection results for four biologically motivated scenarios. All three clustering scenarios detected ($p < 0.001$); negative control correctly not detected.

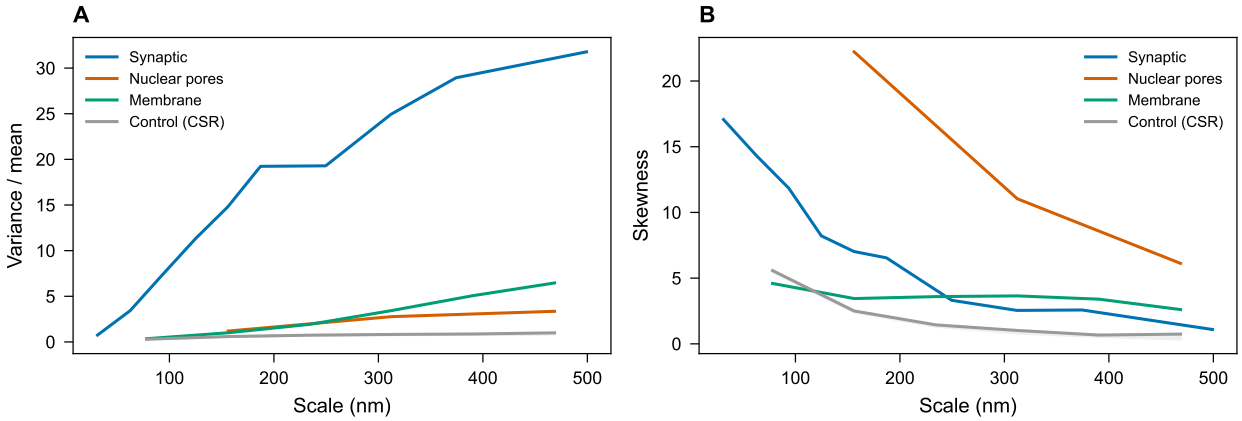

Figure 5: **Multiscale structural fingerprints.** Variance-to-mean ratio curves across spatial scales for different biological structures. Each morphology produces a distinct scale-dependent signature: synaptic receptors show a sharp peak at small scales, nuclear pores produce a peak at intermediate scales reflecting ring geometry, and membrane domains exhibit a broad excess. The mock envelope (gray shading) represents the 95% range from null datasets. These fingerprints encode information about the spatial organization that goes beyond binary detection.